

## DBMS – SELLER AND INVENTORY MANAGEMENT

**Project Title: EcoCart – Sustainable Products E-Commerce Order Management System**

**Skill Task: DBMS**

**Week: 3**

### 1. Database Design

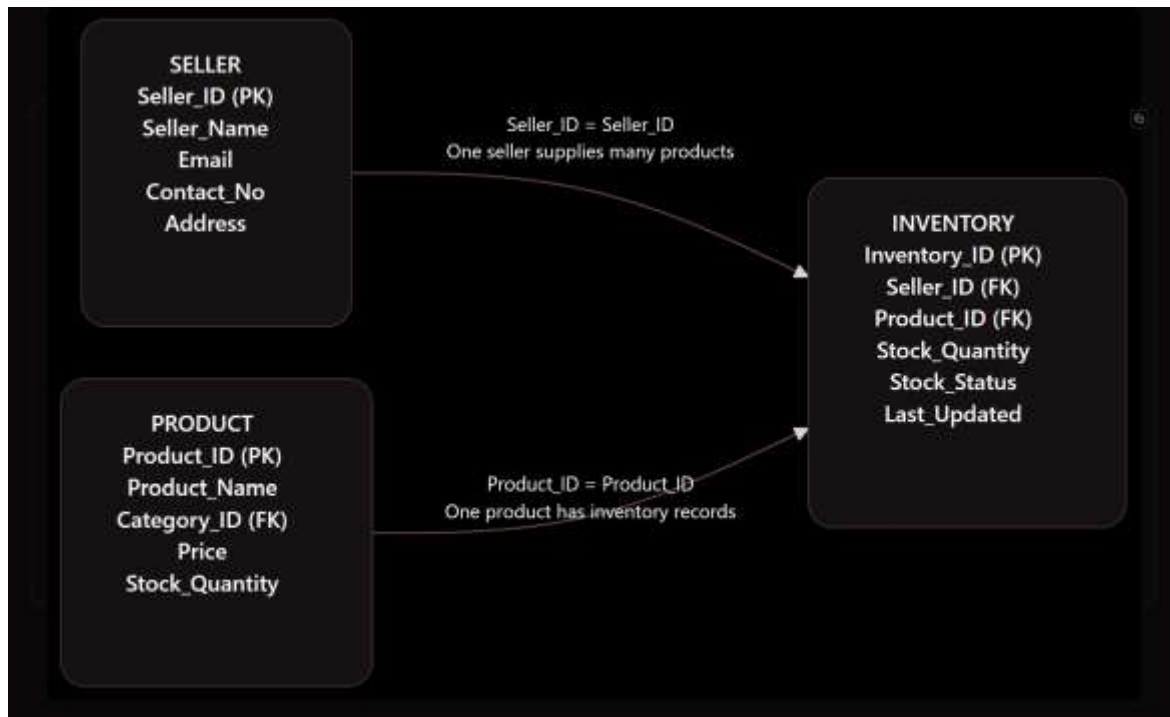
#### 1.1 Seller Table

| <i>Attribute</i>   | <i>Data Type</i> | <i>Constraint</i> |
|--------------------|------------------|-------------------|
| <i>Seller_ID</i>   | INT              | Primary Key       |
| <i>Seller_Name</i> | VARCHAR(100)     | NOT NULL          |
| <i>Email</i>       | VARCHAR(100)     | NOT NULL, UNIQUE  |
| <i>Contact_No</i>  | VARCHAR(15)      | NOT NULL, UNIQUE  |
| <i>Address</i>     | VARCHAR(200)     | NOT NULL          |

#### 1.2 Inventory Table

| <i>Attribute</i>      | <i>Data Type</i> | <i>Constraint</i> |
|-----------------------|------------------|-------------------|
| <i>Inventory_ID</i>   | INT              | Primary Key       |
| <i>Product_ID</i>     | INT              | Foreign Key       |
| <i>Seller_ID</i>      | INT              | Foreign Key       |
| <i>Stock_Quantity</i> | INT              | NOT NULL          |
| <i>Stock_Status</i>   | VARCHAR(20)      | NOT NULL          |
| <i>Last_Updated</i>   | DATE             | NOT NULL          |

### 2. Table Relationships



### 3. Table Creation

#### 3.1 Create Seller Table

```

SQL> CREATE TABLE Seller (
2     Seller_ID INT PRIMARY KEY,
3     Seller_Name VARCHAR(100) NOT NULL,
4     Email VARCHAR(100) NOT NULL UNIQUE,
5     Contact_No VARCHAR(15) NOT NULL UNIQUE,
6     Address VARCHAR(200) NOT NULL
7 );
    
```

Table created.

#### 3.2 Create Inventory Table

```

SQL> CREATE TABLE Inventory (
2     Inventory_ID INT PRIMARY KEY,
3     Product_ID INT NOT NULL,
4     Seller_ID INT NOT NULL,
5     Stock_Quantity INT NOT NULL,
6     Stock_Status VARCHAR(20) NOT NULL,
7     Last_Updated DATE NOT NULL,
8     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID),
9     FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID)
10 );
    
```

Table created.

### 4. Seller Data Insertion

```
SQL> INSERT INTO Seller
  2 VALUES (202, 'Organic World Traders', 'organicworld@gmail.com',
  3          '90000000002', 'Coimbatore');

1 row created.

SQL> INSERT INTO Seller
  2 VALUES (203, 'Nature Basket Suppliers', 'naturebasket@gmail.com',
  3          '90000000003', 'Madurai');

1 row created.

SQL> INSERT INTO Seller
  2 VALUES (204, 'Eco Living Products', 'ecoliving@gmail.com',
  3          '90000000004', 'Bangalore');

1 row created.

SQL> INSERT INTO Seller
  2 VALUES (205, 'Pure Earth Organics', 'pureearth@gmail.com',
  3          '90000000005', 'Salem');

1 row created.
```

## 5. Inventory Data Insertion

```
SQL> INSERT INTO Inventory
  2 VALUES (301, 101, 201, 45, 'Available', '01-SEP-2026');

1 row created.

SQL> INSERT INTO Inventory
  2 VALUES (302, 102, 201, 40, 'Available', '01-SEP-2026');

1 row created.

SQL> INSERT INTO Inventory
  2 VALUES (303, 103, 202, 25, 'Available', '02-SEP-2026');

1 row created.

SQL> INSERT INTO Inventory
  2 VALUES (304, 104, 204, 0, 'Unavailable', '02-SEP-2026');

1 row created.
```

## 6. Seller Product Information

### 6.1 Display Seller and Product Details

This query combines Seller, Inventory, and Product information.

```
SQL> SELECT
  2     s.Seller_Name,
  3     p.Product_Name,
  4     i.Stock_Quantity,
  5     i.Stock_Status
  6 FROM Seller s
  7 JOIN Inventory i
  8 ON s.Seller_ID = i.Seller_ID
  9 JOIN Product p
 10 ON i.Product_ID = p.Product_ID
 11 ORDER BY s.Seller_Name;
```

SELLER\_NAME

PRODUCT\_NAME

STOCK\_QUANTITY STOCK\_STATUS

Eco Living Products  
Bamboo Storage Box  
0 Unavailable

Green Harvest Suppliers  
Black Pepper  
40 Available

SELLER\_NAME

PRODUCT\_NAME

STOCK\_QUANTITY STOCK\_STATUS

Green Harvest Suppliers  
Organic Turmeric Powder  
45 Available

## 7. Inventory Status

### 7.1 Display Available Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Stock_Quantity,
  5     i.Stock_Status
  6 FROM Product p
  7 JOIN Inventory i
  8 ON p.Product_ID = i.Product_ID
  9 WHERE i.Stock_Status = 'Available';
```

PRODUCT\_ID

PRODUCT\_NAME

STOCK\_QUANTITY STOCK\_STATUS

```
-----
      101
Organic Turmeric Powder
      45 Available
```

```
      102
Black Pepper
      40 Available
```

PRODUCT\_ID

PRODUCT\_NAME

STOCK\_QUANTITY STOCK\_STATUS

```
-----
      103
Organic Honey
      25 Available
```

## 7.2 Display Unavailable Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Stock_Quantity,
  5     i.Stock_Status
  6 FROM Product p
  7 JOIN Inventory i
  8 ON p.Product_ID = i.Product_ID
  9 WHERE i.Stock_Status = 'Unavailable';
```

PRODUCT\_ID

PRODUCT\_NAME

STOCK\_QUANTITY STOCK\_STATUS

```
-----
      104
Bamboo Storage Box
      0 Unavailable
```

## 8. Inventory Update

Demonstrate updating stock information.

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Stock_Quantity,
  5     i.Stock_Status
  6 FROM Product p
  7 JOIN Inventory i
  8 ON p.Product_ID = i.Product_ID
  9 WHERE i.Stock_Status = 'Unavailable';
```

PRODUCT\_ID

PRODUCT\_NAME

STOCK\_QUANTITY STOCK\_STATUS

```
-----
      104
Bamboo Storage Box
      0 Unavailable
```

```
SQL> UPDATE Inventory
  2 SET Stock_Quantity = 30,
  3     Stock_Status = 'Available',
  4     Last_Updated = '04-SEP-2026'
  5 WHERE Inventory_ID = 303;
```

1 row updated.

## 9. Inventory Status Reports

### 9.1 Complete Inventory Report

```
SQL> SELECT
  2     i.Inventory_ID,
  3     s.Seller_Name,
  4     p.Product_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status,
  7     i.Last_Updated
  8 FROM Inventory i
  9 JOIN Seller s
 10 ON i.Seller_ID = s.Seller_ID
 11 JOIN Product p
 12 ON i.Product_ID = p.Product_ID
 13 ORDER BY i.Inventory_ID;
```



|                         |              |           |
|-------------------------|--------------|-----------|
| INVENTORY_ID            |              |           |
| -----                   |              |           |
| SELLER_NAME             |              |           |
| -----                   |              |           |
| PRODUCT_NAME            |              |           |
| -----                   |              |           |
| STOCK_QUANTITY          | STOCK_STATUS | LAST_UPDA |
| -----                   |              |           |
| 301                     |              |           |
| Green Harvest Suppliers |              |           |
| Organic Turmeric Powder |              |           |
| 45 Available            |              | 01-SEP-26 |

|                         |              |           |
|-------------------------|--------------|-----------|
| INVENTORY_ID            |              |           |
| -----                   |              |           |
| SELLER_NAME             |              |           |
| -----                   |              |           |
| PRODUCT_NAME            |              |           |
| -----                   |              |           |
| STOCK_QUANTITY          | STOCK_STATUS | LAST_UPDA |
| -----                   |              |           |
| 302                     |              |           |
| Green Harvest Suppliers |              |           |
| Black Pepper            |              |           |
| 40 Available            |              | 01-SEP-26 |

|                |              |           |
|----------------|--------------|-----------|
| INVENTORY_ID   |              |           |
| -----          |              |           |
| SELLER_NAME    |              |           |
| -----          |              |           |
| PRODUCT_NAME   |              |           |
| -----          |              |           |
| STOCK_QUANTITY | STOCK_STATUS | LAST_UPDA |
| -----          |              |           |

## 9.2 Available and Unavailable Product Count

```
SQL> SELECT
2      Stock_Status,
3      COUNT(*) AS Total_Products
4  FROM Inventory
5  GROUP BY Stock_Status;
```

|              |                |
|--------------|----------------|
| STOCK_STATUS | TOTAL_PRODUCTS |
| -----        | -----          |
| Available    | 3              |
| Unavailable  | 1              |

## 9.3 Total Stock by Seller

```
SQL> SELECT
2     s.Seller_Name,
3     SUM(i.Stock_Quantity) AS Total_Stock
4 FROM Seller s
5 JOIN Inventory i
6 ON s.Seller_ID = i.Seller_ID
7 GROUP BY s.Seller_Name
8 ORDER BY s.Seller_Name;
```

SELLER\_NAME

---

TOTAL\_STOCK

---

Eco Living Products  
0

Green Harvest Suppliers  
85

Organic World Traders  
30

## 9.4 Products with Zero Stock

This is particularly useful for demonstrating unavailable products.

```
SQL> SELECT
2     p.Product_ID,
3     p.Product_Name,
4     s.Seller_Name,
5     i.Stock_Quantity,
6     i.Stock_Status
7 FROM Product p
8 JOIN Inventory i
9 ON p.Product_ID = i.Product_ID
10 JOIN Seller s
11 ON i.Seller_ID = s.Seller_ID
12 WHERE i.Stock_Quantity = 0;
```

PRODUCT\_ID

---

PRODUCT\_NAME

---

SELLER\_NAME

---

STOCK\_QUANTITY STOCK\_STATUS

---

104  
Bamboo Storage Box  
Eco Living Products  
0 Unavailable